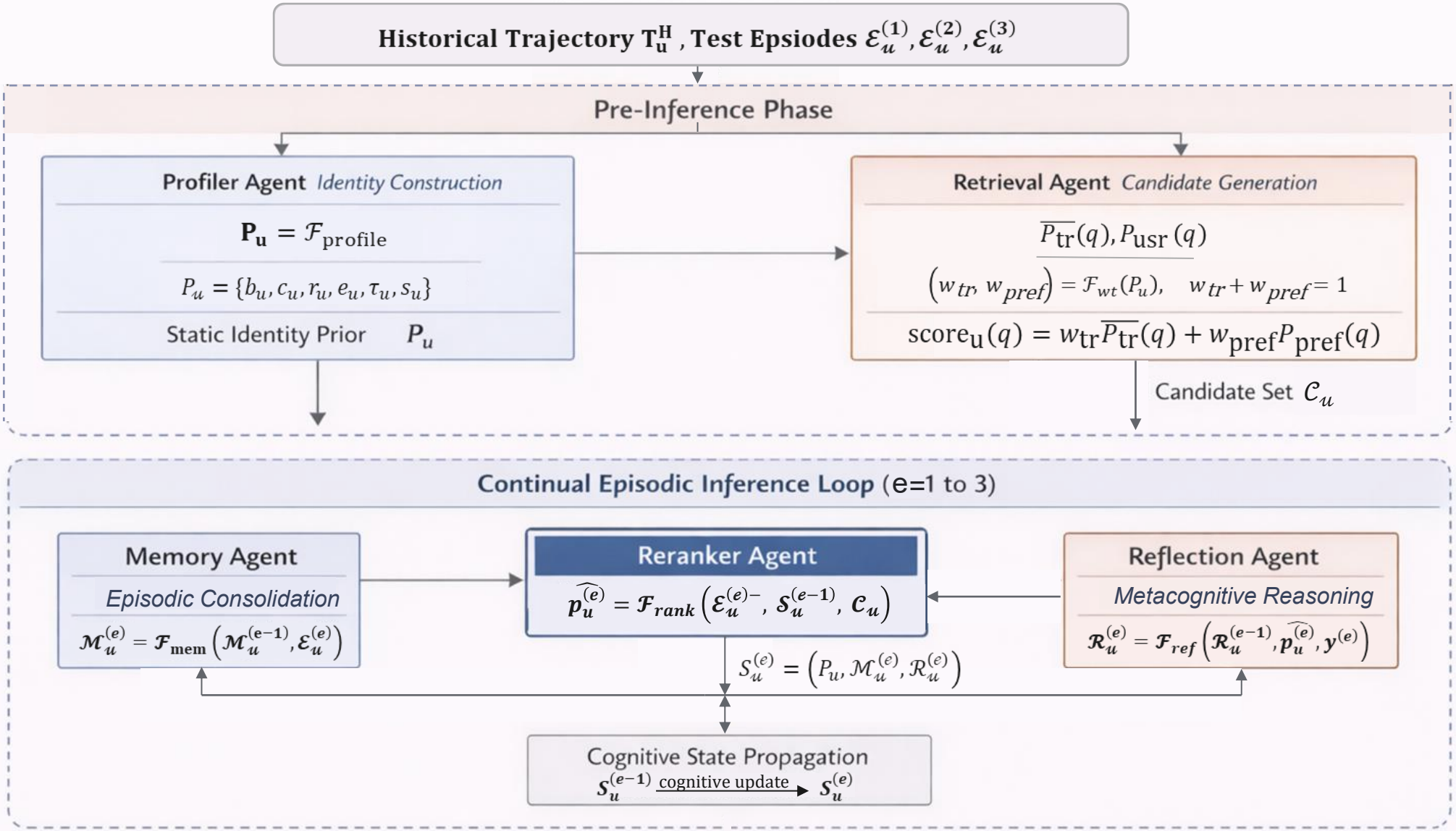
Historical Trajectory T_u^H , Test Epsiodes $\mathcal{E}_u^{(1)}, \mathcal{E}_u^{(2)}, \mathcal{E}_u^{(3)}$
Pre-Inference Phase
Profiler Agent Identity Construction
$\mathbf{P_u} = \mathcal{F}_{\text{profile}}$
$P_u = \{b_u, c_u, r_u, e_u, \tau_u, s_u\}$
Static Identity Prior P_u
Retrieval Agent Candidate Generation
$\overline{P_{\text{tr}}}(q), P_{\text{usr}}(q)$
$(w_{tr}, w_{pref}) = \mathcal{F}_{wt}(P_u), \quad w_{tr} + w_{pref} = 1$
$\text{score}_u(q) = w_{\text{tr}}\overline{P_{\text{tr}}}(q) + w_{\text{pref}}P_{\text{pref}}(q)$
Candidate Set $\mathcal{C}_u$
Continual Episodic Inference Loop (e=1 to 3)
Memory Agent
Episodic Consolidation
$\mathcal{M}_u^{(e)} = \mathcal{F}_{\text{mem}}\left(\mathcal{M}_u^{(e-1)}, \mathcal{E}_u^{(e)}\right)$
Reranker Agent
$\widehat{p_u^{(e)}} = \mathcal{F}_{rank}\left(\mathcal{E}_u^{(e)-}, \mathcal{S}_u^{(e-1)}, \mathcal{C}_u\right)$
Reflection Agent
Metacognitive Reasoning
$\mathcal{R}_u^{(e)} = \mathcal{F}_{ref}\left(\mathcal{R}_u^{(e-1)}, \widehat{p_u^{(e)}}, y^{(e)}\right)$
$S_u^{(e)} = \left(P_u, \mathcal{M}_u^{(e)}, \mathcal{R}_u^{(e)}\right)$
Cognitive State Propagation
$\mathcal{S}_u^{(e-1)} \xrightarrow{\text{cognitive update}} \mathcal{S}_u^{(e)}$